

SE Practices 1 | SEP1

Summary

CONTENTS

1. Object-Oriented Analysis and Design

2. Unified Modeling Language (UML)

3. Unified Process (UP)

4. Iterative Development

5. Domain Models

2

2

2

2

2

1. OBJECT-ORIENTED ANALYSIS AND DESIGN

“Responsibility-driven design”

- 1) Requirements analysis: What should the product do?
 - Software Requirements Specification (SRS)
- 2) Domain analysis: What is the problem domain?
 - Static View: Domain Model (UML notation: Class Diagram)
 - Dynamic View: Behavioural Models (UML notation: Interaction Diagram, State Machine Diagram, Activity Diagram, etc. depending on the domain)

TODO:

diagrams oo slides 6

TODO:

oo slides in general

Analysis: Investigation of the objects in the problem domain

Design: Defining software objects and how they collaborate to fulfill the requirements

- 1) Define Use Cases
- 2) Define a Domain Model
- 3) Assign Object Responsibilities and Draw Interaction Diagrams
- 4) Define Design Class Diagrams

2. UNIFIED MODELING LANGUAGE (UML)

Visual language for specifying, constructing and documenting the artifacts of systems.

Perspectives to apply UML (from abstract to specific): Conceptual, Specification, Implementation

UML Notation for Domain Models:

- Classes for sets of similar objects in the problem domain
- Attributes for properties of those objects
- Associations for relationships between those objects
- Generalisations in between classes with a “each X is a Y” relationship

3. UNIFIED PROCESS (UP)

An iterative software development process for building object-oriented systems.

Software development process: describes an approach to building, deploying and possibly maintaining software.

4. ITERATIVE DEVELOPMENT

5. DOMAIN MODELS